A Certified Stationary-Orbit Bifurcation for a Cone-Volume-Weighted Radial–Normal Angle on Prolate Spheroids

Katsushi Furuta
Independent Researcher

September 5, 2026

Abstract

We study a cone-volume-weighted radial–normal angle functional with the base point as the variable and prove that its stationary set changes its local component structure along the one-parameter family of prolate spheroids. Rotational symmetry forces the center to remain stationary, while on one side of a critical axis ratio there is a stationary circle in the equatorial plane that shrinks into the center at the critical parameter. A local analytic reduction expresses the bifurcation through an equatorial quadratic coefficient $Q(a)$, a quartic coefficient $H_4(a)$, and an axial quadratic coefficient $Q_z(a)$. Rigorous Arb/Acb ball arithmetic certifies a unique simple zero

$$4.72438340452113340672 < a_c < 4.72438340452113340673,$$

with $Q'(a_c) < 0$, $H_4(a_c) < 0$, and $Q_z(a) > 0$ on $[4.7, 4.75]$. The resulting stationary circle satisfies $r(a) \to 0$ as $a \uparrow a_c$. The computer-assisted part is reproducible from a public twelve-object, SHA-256-pinned certificate chain; symbolic audits independently connect the analytic kernel $h(x) = \arccos^2 x$ to the regularized ${}_0F_1$ derivative formulas and to the raw axial kernel. Thus the experimental discovery, analytic reduction, and rigorous interval proof are separated explicitly.

Keywords. stationary orbit; bifurcation; prolate spheroid; cone-volume measure; radial–normal angle; interval arithmetic; computer-assisted proof; experimental mathematics.

1 Introduction

We treat the base point of a geometric functional as a variable and ask how its stationary set changes when the underlying body is deformed. Classical center constructions usually select a point. Once uniqueness is lost, however, continuous symmetry can force positive-dimensional stationary sets. The present example makes this mechanism explicit and rigorous.

For a prolate spheroid of polar-to-equatorial axis ratio a, numerical exploration first indicated an equatorial stationary circle that contracts toward the center as a approaches a value near 4.72438. The experiment suggested a local bifurcation governed by a sign change of the transverse quadratic coefficient. We then replaced the exploratory computation by a computer-assisted proof: symbolic kernel audits, rigorous ball-arithmetic integration, a 20-decimal endpoint sign certificate, and a separate axial positivity certificate. This progression from experiment to theorem is the main computational narrative of the paper and is consistent with the role of reproducible computer experiments in experimental mathematics [9].

The result is local. We prove the existence and local uniqueness of the stationary circle near the center and the change in the local stationary-set labels. We do not claim a global enumeration of every stationary component for every axis ratio.

The logical order is deliberately narrow: the main theorem is the existence of a local basepoint bifurcation; the stationary circle is the geometric carrier of that theorem; and interval certification of the local coefficients is the device that converts the observed pattern into a proof.

2 Basepoint bifurcation

Let P be a basepoint space, Λ a parameter space, and $E_\lambda : P \to \mathbb{R}$ a family of functionals. Set

$$\mathcal{S} = \{(\lambda, p) : d_p E_\lambda = 0\}, \qquad \pi : \mathcal{S} \to \Lambda.$$

We label each local stationary component by its dimension, normal Morse index, nullity, and isotropy type. For a positive-dimensional component we use the normal Morse index rather than the ordinary Morse index because the stationary circle is a Morse–Bott critical manifold; its tangential zero mode is forced by symmetry.

Definition 2.1 (Local basepoint bifurcation). *A point $(\lambda_0, p_0) \in \mathcal{S}$ is a local basepoint bifurcation point if, for every sufficiently small two-sided neighborhood $I \ni \lambda_0$ and neighborhood $U \ni p_0$, the restricted projection $\pi : \mathcal{S} \cap (I \times U) \to I$ admits no local trivialization preserving these labels. Here a label-preserving local trivialization means a homeomorphism $\Phi : \mathcal{S} \cap (I \times U) \to I \times F$ with $\mathrm{pr}_I \circ \Phi = \pi$ such that corresponding connected components in the fibers carry the same component-dimension, normal-index, nullity, and isotropy labels.*

This definition concerns the germ of the stationary set and therefore does not require a global classification away from (λ_0, p_0).

3 Cone-volume-weighted radial–normal angle

Let $K \subset \mathbb{R}^3$ be a smooth convex body and $p \in \operatorname{int} K$. For $x \in \partial K$ with outward unit normal $\nu_K(x)$ define

$$\alpha_{K,p}(x) = \arccos\left(\frac{(x-p)\cdot \nu_K(x)}{|x-p|}\right), \qquad d\mu_{K,p}(x) = \frac{(x-p)\cdot \nu_K(x)}{3\,\mathrm{Vol}(K)}\, dA(x),$$

and

$$E_K(p) = \int_{\partial K} \alpha_{K,p}(x)^2\, d\mu_{K,p}(x).$$

The divergence theorem gives $\int_{\partial K} (x-p)\cdot \nu_K\, dA = 3\,\mathrm{Vol}(K)$, so $\mu_{K,p}$ is a probability measure. Cone-volume measure is natural in convex geometry [1, 2, 3]; related center constructions include radial and generalized centers [5, 6, 7, 8].

Proposition 3.1 (Similarity invariance). *If $T(x) = sUx + q$ with $s > 0$ and $U \in O(3)$, then $E_{T(K)}(T(p)) = E_K(p)$.*

The angle is similarity invariant, while the support-distance factor, area element, and volume scale by s, s^2, and s^3, respectively. Hence the normalized measure is invariant. In particular, signs, Morse indices, and nullities of the basepoint Hessian are unchanged by uniform rescaling. We therefore normalize the equatorial semiaxis to one below.

4 Prolate spheroids and local reduction

Consider

$$K_a = \{(x, y, z) : x^2 + y^2 + z^2/a^2 \le 1\}, \qquad a \ge 1.$$

The full isometry group is $D_{\infty h}$. In particular it contains the $O(2)$ action about the z axis and the equatorial reflection $\sigma_h : (x, y, z) \mapsto (x, y, -z)$. The center is fixed by the full group and is therefore stationary for every a. On $z = 0$, reflection symmetry forces $\partial_z E_a = 0$, and the remaining equation depends only on $r = (x^2 + y^2)^{1/2}$. Thus a nonzero radial solution on the equatorial plane is already a three-dimensional stationary point, and the axial $O(2)$ symmetry turns one such point into an entire stationary circle.

Near the center,

$$E_a(r) = E_a(0) + \frac{Q(a)}{2} r^2 + \frac{H_4(a)}{24} r^4 + O(r^6),$$

so

$$\partial_r E_a(r) = rG(a, r^2), \qquad G(a, s) = Q(a) + \frac{H_4(a)}{6} s + O(s^2).$$

Lemma 4.1 (Real analyticity near the center)**.** *For a in a neighborhood of $[4.7, 4.75]$ and p sufficiently close to the center, $E_a(p)$ is real analytic in (a, p).*

Indeed $h(x) = \arccos^2 x$ has a removable analytic endpoint at $x = 1$,

$$h(x) = 2(1 - x) + \frac{1}{3}(1 - x)^2 + O((1 - x)^3),$$

while convexity and compactness of the parameter neighborhood keep the radial–normal cosine uniformly separated from -1. The geometric factors in the boundary parametrization are analytic, and on the compact angular domain their derivatives admit uniform bounds. Hence the local power series may be integrated term by term, giving analytic dependence on (a, p). In particular the even radial expansion above, the factorization $\partial_r E_a = rG(a, r^2)$, and the Hadamard factorization used below are justified by this lemma.

4.1 Canonical coefficient kernels

For reproducibility, the interval certificates enclose the following one-dimensional kernels directly. Put $s = \sin\theta$, $c_\theta = \cos\theta$,

$$\ell = s^2 + a^2 c_\theta^2, \qquad w = (a^2 s^2 + c_\theta^2)^{1/2}, \qquad C = \frac{a}{w\sqrt{\ell}},$$

and write $h_j = h^{(j)}(C)$ for $h(x) = \arccos^2 x$. The transverse quadratic kernel is

$$B_2 = \frac{C}{\ell^2}\left\{Ch_2(\ell - 1)^2 s^2 + h_1\left[(2\ell^2 - 4\ell + 3)s^2 - 2\ell\right]\right\}, \qquad Q(a) = \frac{1}{2}\int_0^{\pi/2} sB_2 \, d\theta.$$

For the fourth derivative, define

$$\begin{aligned}A_4 = \frac{C}{\ell^4}\{&C^3 h_4(\ell-1)^4\\ &+C^2h_3(4\ell^4-24\ell^3+54\ell^2-52\ell+18)\\ &+Ch_2(-24\ell^3+108\ell^2-168\ell+87)\\ &+h_1(36\ell^2-120\ell+105)\},\\ A_2 = -\frac{6C}{\ell^3}\{&C^2h_3(\ell-1)^2+Ch_2(4\ell^2-12\ell+9)\\ &+h_1(2\ell^2-12\ell+15)\},\end{aligned}$$

$$A_0 = \frac{3C}{\ell^2}(Ch_2+3h_1), \qquad B_4 = \frac{3}{4}A_4 s^4 + A_2 s^2 + 2A_0,$$

so that $H_4(a) = \frac{1}{2}\int_0^{\pi/2} sB_4\,d\theta$. For the axial certificate put $v = ac_\theta$, $w_z = c_\theta/a$, and $d = v/\ell - w_z$. The audited raw second derivative is

$$F_z''(0) = C^2h_2d^2 + Ch_1\left(-\frac{1}{\ell}+\frac{3v^2}{\ell^2}-\frac{4w_zv}{\ell}+2w_z^2\right), \qquad Q_z(a) = \int_0^{\pi/2} sF_z''(0)\,d\theta.$$

These are exactly the kernels used by the SHA-pinned symbolic audits and Arb/Acb certificates described in Section 10.

5 Experimental discovery and certification design

Exploratory high-precision computations were used only to locate the candidate sign change and to identify the coefficients that should control the local branch. They are not proof objects. The proof stage fixes closed parameter intervals, analytic formulas, outward-rounded Arb/Acb arithmetic, and fail-closed sign conditions.

The certificate architecture separates three questions. First, symbolic audits check the equatorial quadratic and quartic kernels. Second, a rigorous interval certificate establishes the sign change and monotonicity of Q and the negativity of H_4. Third, the axial calculation is independently audited and certified, preventing an equatorial stationary point from escaping the three-dimensional stationary condition. A separate symbolic audit verifies the regularized ${}_0F_1$ derivative chain used by the interval kernels.

6 Certified critical axis ratio

Proposition 6.1 (Certified local coefficients)**.** *On* $[4.7, 4.75]$ *there is a unique* a_c *satisfying* $Q(a_c) = 0$*, and*

$$4.72438340452113340672 < a_c < 4.72438340452113340673.$$

Moreover $Q'(a_c) < 0$ *and* $H_4(a_c) < 0$*.*

The endpoint enclosures are

$$\begin{aligned}Q(4.72438340452113340672) \in [&4.459501177799009170989867 72,\\ &4.45950117779900917098986968] \times 10^{-22},\\ Q(4.72438340452113340673) \in [&-5.00431892402172225190109006,\\ &-5.00431892402172225190108810] \times 10^{-22}.\end{aligned}$$

On the five subintervals $[4.70, 4.71], \ldots, [4.74, 4.75]$, the certified enclosures for Q' are respectively

$$[-0.10545, -0.08573], \quad [-0.10492, -0.08527], \quad [-0.10438, -0.08480],$$
$$[-0.10386, -0.08434], \quad [-0.10333, -0.08388],$$

and those for H_4 are

$$[-1.49670, -1.39252], \quad [-1.49333, -1.38959], \quad [-1.48998, -1.38666],$$
$$[-1.48664, -1.38375], \quad [-1.48331, -1.38085].$$

Thus the zero is simple and unique on the stated interval.

The analytic implicit-function theorem [4] applied to $G(a, s) = 0$ gives a unique local branch $s(a)$ with $s(a_c) = 0$ and

$$s'(a_c) = -\frac{6Q'(a_c)}{H_4(a_c)} < 0.$$

Hence $s(a) > 0$ for $a < a_c$ sufficiently close to a_c, whereas $s(a) < 0$ on the opposite side, and the stationary radius $r(a) = \sqrt{s(a)}$ satisfies $r(a) \to 0$ as $a \uparrow a_c$. More precisely,

$$r(a)^2 = s(a) = -\frac{6Q(a)}{H_4(a_c)} + O((a - a_c)^2).$$

The coefficient $c = 6Q'(a_c)/H_4(a_c)$ is rigorously enclosed by $[0.3414, 0.4517]$; the value 0.39474123 is only a non-certified reference-point estimate.

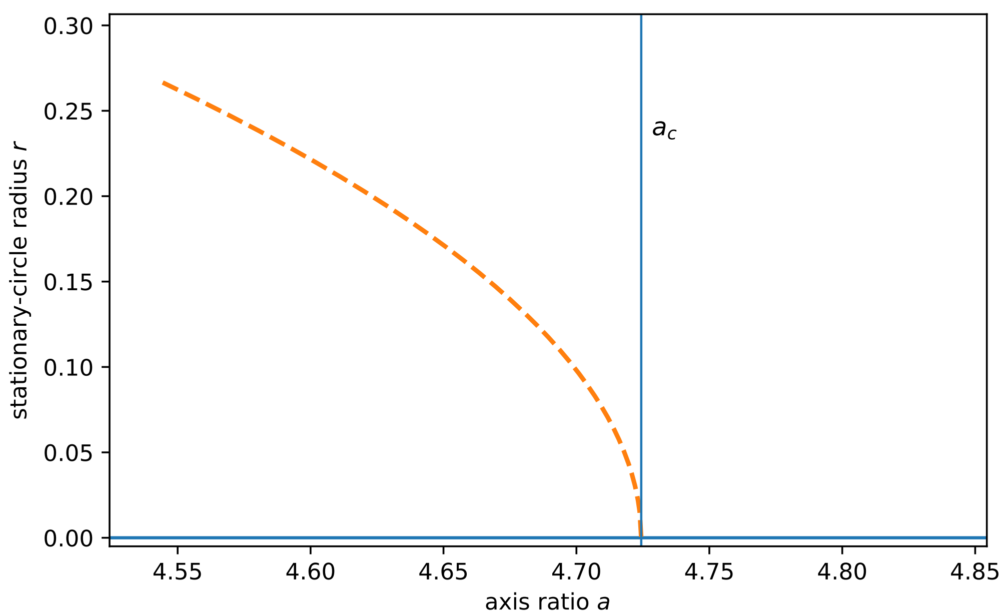

Figure 1: Certified local bifurcation picture. The dashed curve represents the leading law $r_*(a)^2 = c(a_c - a)$. The coefficient satisfies the certified enclosure $c \in [0.3414, 0.4517]$; 0.39474123 is a non-certified reference-point estimate. The figure is schematic, not a global numerical continuation.

7 Axial confinement and the stationary circle

Write the axial derivative near the center as

$$\partial_z E_a(x, y, z) = zH(a, x, y, z^2).$$

Real analyticity implies continuity of H. The certified axial coefficient satisfies

$$Q_z(a) > 0 \quad \text{for all } a \in [4.7, 4.75],$$

with rigorous lower bound 0.0885587746621582. Since $H(a_c, 0, 0, 0) = Q_z(a_c) > 0$, continuity gives $H > 0$ in a sufficiently small neighborhood of $(a_c, 0)$. Thus $\partial_z E_a = 0$ forces $z = 0$ there.

Consequently the nonzero equatorial solution produces a stationary circle $C_a \cong S^1$. The radial normal eigenvalue is negative. The axial eigenvalue remains positive on C_a for $a < a_c$ sufficiently close to a_c, by $Q_z(a_c) > 0$, $r(a) \to 0$, and continuity of the Hessian. The tangential zero eigenvalue is forced by rotational symmetry. Hence C_a is locally Morse–Bott.

Across the critical value the germ of the stationary set therefore changes from "isolated center plus a one-dimensional circle" to "isolated center only." This change in component dimension already obstructs label-preserving local triviality. Independently, at $a = a_c$ the equatorial Hessian eigenvalue vanishes, so the nullity of the center increases by two.

The claim is intentionally local: no assertion is made here that every stationary component is enumerated over the full axis-ratio range. A complete three-dimensional label table belongs to a separate global certification problem.

8 Main theorem

Theorem 8.1 (Certified stationary-orbit bifurcation)**.** *For the cone-volume-weighted radial–normal angle functional on K_a, $(a_c, 0)$ is a local basepoint bifurcation point. In a sufficiently small neighborhood of the center,*

$$\mathrm{Crit}(E_a) = \{0\} \sqcup C_a \quad (a < a_c \textit{ sufficiently close}),$$

where $C_a \cong S^1$ and $r(a) \to 0$ as $a \uparrow a_c$; at $a = a_c$ the equatorial part of the center Hessian has nullity two; and for $a > a_c$ sufficiently close to a_c the only local stationary component is the center. Therefore the local stationary-set projection is not label-preserving locally trivial across a_c.

Proof structure. The argument consists of four steps. First, $D_{\infty h}$ symmetry, and especially the equatorial reflection, reduces the local three-dimensional stationary problem to a radial equation. Second, interval arithmetic certifies the unique simple zero of Q, the sign $Q'(a_c) < 0$, and $H_4(a_c) < 0$. Third, analyticity and even symmetry factor the radial derivative as $\partial_r E_a = rG(a, r^2)$, and the implicit-function theorem yields the unique nonzero local branch and its collapse to the center. Fourth, the independent axial certificate and Hessian continuity close the full stationary condition and identify the Morse–Bott labels. The change of these labels across a_c rules out a label-preserving local trivialization.

9 Endpoint regularization

The direct $\arccos^2$ representation is poorly conditioned for numerical differentiation near an endpoint. Set

$$\beta = \arctan((a - a^{-1}) \sin\theta \cos\theta), \qquad z = \beta^2,$$

and define

$$S = {}_0F_1(;3/2;-z/4), \quad T = {}_0F_1(;5/2;-z/4), \quad U = {}_0F_1(;7/2;-z/4), \quad V = {}_0F_1(;9/2;-z/4).$$

For $h(x) = \arccos^2 x$, the derivatives used by the interval kernels are represented by

$$h_1 = -\frac{2}{S}, \qquad h_2 = \frac{2T}{3S^3},$$

$$h_3 = \frac{2U}{15S^4} - \frac{2T^2}{3S^5}, \qquad h_4 = \frac{2V}{105S^5} - \frac{4UT}{9S^6} + \frac{10T^3}{9S^7}.$$

At $\beta = 0$ these formulas are interpreted by analytic continuation through the entire ${}_0F_1$ representations, with $S(0) = 1$. On the target interval $S = \sin\beta/\beta$ has a positive lower bound, so the denominators remain uniformly separated from zero. This regularization is what permits rigorous resolution of endpoint sign margins of order 5×10^{-22}. The four identities above are themselves checked by an independent symbolic audit, with all four differences simplifying exactly to zero.

10 Reproducibility and proof objects

The proof archive contains twelve SHA-256-pinned objects under `CERTIFICATES/prolate/local_bifurcation/`. They comprise the cap-chain certificate and reports, the symbolic audits for the equatorial kernels, the 20-decimal a_c certificate and JSON report, the raw-kernel Q_z symbolic audit, the Q_z Arb certificate and JSON report, and the ${}_0F_1$ derivative-chain symbolic audit. Running `sha256sum -c SHA256SUMS.txt` gives 12/12 OK at the canonical commit

`abf704b8701f3da8eb41d8a99408570a413b2214`.

The a_c run uses Python 3.13.14, 100 decimal digits, and tolerance 10^{-40}. The Q_z run uses Python 3.13.14, 70 decimal digits, tolerance 10^{-28}, and five parameter subdivisions. Recorded common components include python-flint 0.9.0, FLINT 3.6.0, SymPy 1.14.0, and mpmath 1.3.0.

The Q_z certificate gates the manuscript claim itself: `status=CERTIFIED` requires both positivity on all five parameter intervals and `worst_lower >= 0.0885587746621582`. The symbolic Q_z audit differentiates the raw weighted kernel twice while keeping h abstract; it does not depend on a pre-substituted finite Taylor jet.

10.1 Fail-closed certification

A numerical evaluation is not counted as a proof object merely because it returns a plausible decimal value. The certification stage fixes closed parameter intervals, formulas, precision settings, and outward rounding. If an interval contains zero where a strict sign is required, if an evaluation becomes non-finite, or if a recorded input or proof object fails its integrity check, the run does not emit a certified result. It must either refine the interval or terminate without certification. This is the operational distinction between exploratory numerics and mathematical enclosure used throughout the paper.

10.2 Global continuation is not a premise

The repository also contains machinery for global interval continuation and component tracking. That machinery is useful for a future completeness theorem, but it is not a premise of Theorem 8.1. The theorem here closes with local coefficient certification, analytic bifurcation analysis, and the axial positivity certificate alone.

11 Scope and relation to center theory

The result concerns a local germ, not a global classification. Interval continuation machinery for global component tracking is therefore not a premise of Theorem 8.1. The novelty is also not a new abstract $O(2)$ bifurcation theorem. Rather, it is the rigorous realization of a positive-dimensional stationary orbit and its collapse in a concrete convex-geometric functional, with the experimental and certified stages exposed separately.

Classical work on selectors, radial centers, and generalized centers emphasizes distinguished points [5, 6, 7, 8]. Here the loss of uniqueness is itself informative: symmetry turns an off-axis stationary point into an orbit, and the orbit becomes the geometric carrier of the bifurcation.

The scope can be summarized as follows: this paper proves existence of a certified local bifurcation; a separate continuation problem would prove completeness over a larger parameter domain. Keeping these two claims separate prevents a global computation from becoming an unnecessary premise of the local theorem.

12 Outlook

Breaking rotational symmetry by passing from a prolate spheroid to a nearby triaxial ellipsoid should split the stationary circle into finitely many isolated stationary points. At the symmetry level one expects the transition $D_{\infty h} \to D_{2h}$. Tracking component dimension, normal Morse index, nullity, and isotropy type simultaneously provides a natural way to compare such symmetry-breaking unfoldings with the orbit collapse proved here.

13 Conclusion

A computational experiment on prolate spheroids led to a candidate stationary-circle bifurcation. Symbolic reduction and rigorous interval arithmetic convert that observation into a theorem: a unique critical axis ratio is certified to twenty decimal places, a stationary circle exists on one side and contracts to the center, and axial positivity closes the three-dimensional stationary condition. The public certificate chain makes the numerical part independently repeatable and distinguishes exploratory evidence from proof objects.

Acknowledgments

The author acknowledges the influence of the experimental-mathematics viewpoint developed by Jonathan Borwein and David Bailey [9].

A Certificate checklist

The final proof state used by the manuscript satisfies the following reproducibility checks.

- All twelve proof objects are pinned by `SHA256SUMS.txt`, and `sha256sum -c` reports 12/12 OK.
- The equatorial symbolic audits are

```
prolate_B2_symbolic_audit.py
prolate_H4_symbolic_audit.py.
```

- The derivative-chain audit `prolate_h_chain_symbolic_audit.py` verifies that the symbolic differences between the regularized formulas and the first four derivatives of $h(x) = \arccos^2 x$ simplify identically to zero.

- The raw axial audit `prolate_Qz_symbolic_audit.py` differentiates the weighted kernel twice while retaining the abstract function h.

- The axial interval certificate `prolate_Qz_arb_certificate.py` and its JSON report gate the manuscript lower bound 0.0885587746621582.

- The critical-axis-ratio certificate records the sign conditions for Q, Q', and H_4, the endpoint root bracket, precision settings, and fail-closed behavior.

- Exploratory stage-1 computations are diagnostic only and are not counted as proof objects.

Data and code availability

All computer-assisted proof objects used in this article are publicly archived in the Basepoint Geometry repository [10]. The canonical proof state is commit

`abf704b8701f3da8eb41d8a99408570a413b2214`.

Its local-bifurcation manifest contains twelve SHA-256-pinned objects and verifies 12/12. The separate `bg-prolate-spheroid` repository is diagnostic only and is not evidence for the theorem.

References

[1] K. J. Böröczky and M. Henk, Cone-volume measure of general centered convex bodies, *Adv. Math.* 286 (2016), 703–721.

[2] K. J. Böröczky, E. Lutwak, D. Yang, and G. Zhang, The logarithmic Minkowski problem, *J. Amer. Math. Soc.* 26 (2013), 831–852.

[3] Y. Huang, E. Lutwak, D. Yang, and G. Zhang, Geometric measures in the dual Brunn–Minkowski theory and their associated Minkowski problems, *Acta Math.* 216 (2016), 325–388.

[4] M. Golubitsky, I. Stewart, and D. G. Schaeffer, *Singularities and Groups in Bifurcation Theory*, Vol. II, Springer, 1988.

[5] I. Herburt, M. Moszyńska, and Z. Peradzyński, Remarks on radial centres of convex bodies, *Math. Phys. Anal. Geom.* 8 (2005), 157–172.

[6] M. Moszyńska, Looking for selectors of star bodies, *Geom. Dedicata* 83 (2000), 131–147.

[7] J. O'Hara, Renormalization of potentials and generalized centers, *Adv. Appl. Math.* 48 (2012), 365–392.

[8] S. Sakata, Stationary radial centers and symmetry of convex polytopes, *Colloq. Math.* 159 (2020), 91–106.

[9] J. M. Borwein and D. H. Bailey, *Mathematics by Experiment: Plausible Reasoning in the 21st Century*, A K Peters, 2004.

[10] K. Furuta, Basepoint Geometry: computational certificates for the prolate-spheroid basepoint bifurcation, software and proof archive, GitHub, canonical commit `abf704b8701f3da8eb41d8a99408570a413b2214`, 2026. `https://github.com/cotaxxxx/basepoint-geometry`.